

SEMANTIC SEARCH SYSTEM

TECHNICAL NOTE:

The objective of this assignment is to create a semantic search system that finds document chunks based on meaning instead of just matching words. I built a system that loads documents, splits them into chunks, converts each chunk into an embedding, stores them in ChromaDB, and lets a user ask a question to retrieve the most relevant chunks. The tools that I used to create this system are sentence_transformers (all-MiniLM-L6-v2) it is used to convert text into embeddings, chromadb to store embeddings and metadata. I used Persistent Client, so the data is saved to disk and doesn't disappear when the notebook restarts. Here I used paragraph base chunking, I split each document by paragraph, since it keeps related ideas together and is simple to implement compared to other chunking methods. I compared keyword search method and semantic search method using same both queries. I tested both methods using same queries. In most cases semantic search gave better results than keyword search especially when the query didn't use the exact words found in the documents.

CONCLUSION:

This project helped me understand how embeddings and vector databases work together to build a system that searches by meaning, not just by words.

Semantic search performed better overall, especially for natural, differently worded questions, which is why it's used in real AI systems like chatbots and document search tools.